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BET-C306
SEMESTER EXAMINATION-DECEMBER 2023
CLASS: B.TECH SEMESTER: III
DIGITAL SYSTEM DESIGN

Time: 3 hour Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any ***five*** questions in about 150 words each. Each question carries six marks.
(5 X 6 = 30 Marks)

Question-1: (i) Convert $(235)_{10}$ to $(\ldots)_2$.
(ii) Convert $(B23)_{16}$ to $(\ldots)_8$.

Question-2: (i) Add 42 to -55 using 8 bit 2's compliment arithmetic.
(ii) Add 297.20 to 483.50 in BCD code.

Question-3: Implement the following multiple output combinational logic circuit using a 4X16 decoder.
$F1=\sum m(0,1,4,7,12,14,15)$
$F2=\sum m(1,3,6,9,12)$
$F3=\sum m(2,3,7,10)$
$F4=\sum m(1,3,5)$

Question-4: Design a full adder circuit using basic logic gates. Provide the truth table for the full adder.

Question-5 Implement the following Boolean function with MUX 8:1
$F(A,B,C,D)=\sum m(0,1,3,4,8,9,15)$

Question-6: Discuss the different types of flip-flops. Differentiate synchronous and asynchronous sequential circuits.

Question-7: Convert JK flip flop to SR flip flop.

Question-8: Design and explain parallel-in, serial-out shift register.

Question-9: Draw and explain the circuit diagram for two input CMOS NOR gate.

Question-10: Discuss the concept of fan-in, fan-out, and noise margin in digital circuits. Why are these parameters important in digital design?

F2=xy'z'+x y

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any ***four*** questions in detail. Each question carries 10 marks.

(4 X 10 = 40 Marks)

Question-1: Simplify the following function using 4 variable k-map:

(i) $F = \bar{B}D + \bar{A}B\bar{C} + A\bar{B}C + AB\bar{C}$

(ii) $F = \prod M\ (6,7,8,9)\,.\,d(10,11,12,13,14,15)$

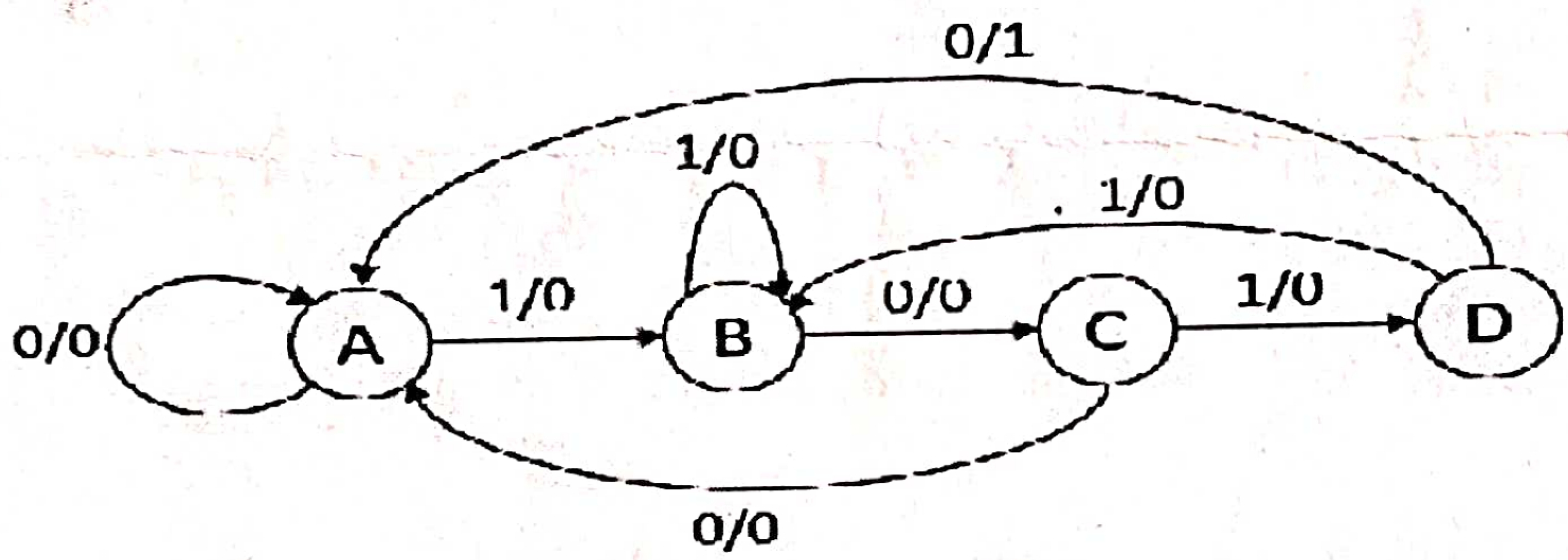
Question-2: Simplify the following function using tabular method:

F(A,B,C,D)= $\sum$m(6,7,8,9,10,11,12,13,14,15)

Question-3: Describe the operation of a 4-bit Magnitude Comparator. How can you cascade two IC -7485s to compare 8 bit binary numbers?

Question-4: Explain 4-bit Parallel adder. Design BCD adder using 74LS83 IC.

Question-5: Design a circuit using SR flip flop that will function as per the state diagram shown below



Question-6: Using T flip-flops:

(a) Design a synchronous counter with the following repeated binary sequence: 0, 3, 5, 6, 0.....

(b) Is the counter self starting? If not than how to make it?

(b) Draw the logic diagram of the counter.

Question-7: Discuss the effect of propagation delay in ripple counters. Design MOD 6 Asynchronous counter using JK flip flop.

Question-8: With the help of a neat diagram, explain the working of two-input TTL NAND gate.